Prescripto AI 2.0 — Testing & Debugging Dossier

Document ID: `DOC-10-TEST`
Classification: Academic Engineering Dossier
Canonical Spec Reference: backend/tests/, docs/TEST_REPORT.md
Date of Extraction: November 4, 2026

---

1. Testing Infrastructure & Methodology

1.1. Test Harness Architecture

- **Testing Framework:** `pytest` (v9.1.1) with `pytest-asyncio` and `anyio`.
- **Execution Setup:** Configured via backend/tests/conftest.py.
- **Dual Persistence Strategy:**
- **SQLite In-Memory Mode (Hermetic Unit Testing):** Default when `TEST_DATABASE_URL` is unset. Fast, isolated, zero-external-dependency execution.
- **PostgreSQL Savepoint Mode (Integration & Concurrency):** When `TEST_DATABASE_URL` is provided, pytest runs `alembic upgrade head` and uses SQLAlchemy's `join_transaction_mode="create_savepoint"`. This allows tests to issue real SQL `COMMIT` statements within worker transactions while rolling back outer savepoints between test functions to preserve test isolation.

---

2. Test Suite Categorization

```
backend/tests/
├── contract/            # Contract parity between OpenAPI schema and FastAPI routes
├── eval_set/            # Synthetic benchmark image dataset and ground-truth JSON
├── integration/         # Multi-component workflow and database integration suites
└── unit/                # Pure domain, cryptographic, and mathematical rule suites
```

2.1. Suite Summary

- **Total Executed Automated Test Cases:** 197
- **Unit Tests:** 38 tests covering pure algorithms, uncertainty models, and cryptography.
- **Integration Tests:** 152 tests covering REST routes, worker concurrency, and S3 storage.
- **Contract Tests:** 7 tests verifying schema parity against `OPENAPI.yaml`.
- **PostgreSQL Concurrency Tests:** 6 tests evaluating simultaneous multi-worker claiming.

---

3. Verified Test Case Matrix

The following matrix documents real test cases executed in the Prescripto AI 2.0 codebase:

Test ID	Module	Scenario Tested	Expected Result	Actual Result	Verification Status
TC-SEC-01	`auth.crypto`	Hash plaintext password and verify against hash	Matching password returns `True`; incorrect password returns `False`	Pass	VERIFIED IN TEST (`test_auth_crypto.py`)
TC-SEC-02	`auth.crypto`	Issue RS256 token and decode with public key	Valid payload returned with claims `sub`, `role`, `exp`	Pass	VERIFIED IN TEST (`test_auth_crypto.py`)
TC-SEC-03	`auth.crypto`	Decode expired RS256 JWT token	Raises `jwt.ExpiredSignatureError`	Pass	VERIFIED IN TEST (`test_auth_crypto.py`)
TC-SEC-04	`audit.logger`	Structlog event with unwhitelisted PHI keys	Strips unapproved keys, preserves `PERMITTED_LOG_KEYS`, records dropped count	Pass	VERIFIED IN TEST (`test_logger.py`)
TC-SEC-05	`storage.client`	Generate and verify HMAC presigned storage URL	Valid before expiration; returns `False` after expiration or tampering	Pass	VERIFIED IN TEST (`test_storage_client.py`)
TC-SEC-06	`storage.client`	Storage key containing path traversal `../`	Intercepted and raises `StorageObjectNotFoundException`	Pass	VERIFIED IN TEST (`test_storage_client.py`)
TC-SAFE-01	`domain.safety`	Same canonical drug prescribed twice on same document	Emits `DUPLICATE_MEDICATION` finding with `FindingStatus.POTENTIAL`	Pass	VERIFIED IN TEST (`test_clinical_safety_rules.py`)
TC-SAFE-02	`domain.safety`	Two distinct brands sharing active ingredient (*Dolo + Combiflam*)	Flags active ingredient overlap (*Paracetamol*)	Pass	VERIFIED IN TEST (`test_clinical_safety_rules.py`)
TC-SAFE-03	`domain.safety`	Unresolved medication line evaluated for duplicates	Emits `NOT_EVALUATED` finding with reason `UNRESOLVED_IDENTITY`	Pass	VERIFIED IN TEST (`test_clinical_safety_rules.py`)
TC-SAFE-04	`domain.safety`	Prescribed dose exceeds ceiling (Paracetamol 1500mg)	Emits `ADVERSE_EFFECT` finding requiring clinical review	Pass	VERIFIED IN TEST (`test_enhanced_safety.py`)
TC-SAFE-05	`domain.safety`	Patient with penicillin allergy prescribed Augmentin	Emits `ALLERGY_CONFLICT` finding flagging contraindication	Pass	VERIFIED IN TEST (`test_enhanced_safety.py`)
TC-UNC-01	`domain.uncertainty`	Raw confidence score 0.45 assigned to extracted field	Mapped to `FieldState.UNREADABLE`	Pass	VERIFIED IN TEST (`test_domain_uncertainty.py`)
TC-UNC-02	`domain.uncertainty`	Unreadable field passed to downstream safety rule	Field is masked and excluded from safety conclusion	Pass	VERIFIED IN TEST (`test_domain_uncertainty.py`)
TC-ING-01	`api.prescriptions`	Upload image without `Idempotency-Key` header	Returns HTTP `400 Bad Request` (`MISSING_IDEMPOTENCY_KEY`)	Pass	VERIFIED IN TEST (`test_prescriptions_api.py`)

Test ID	Module	Scenario Tested	Expected Result	Actual Result	Verification Status
TC-ING-02	`api.prescriptions`	Replay identical upload with same idempotency key	Returns HTTP `202 Accepted` with existing IDs (0 duplicate jobs)	Pass	VERIFIED IN TEST (`test_prescriptions_api.py`)
TC-ING-03	`api.prescriptions`	Replay same key with different file payload	Returns HTTP `409 Conflict` (`IDEMPOTENCY_CONFLICT`)	Pass	VERIFIED IN TEST (`test_prescriptions_api.py`)
TC-WRK-01	`worker.queue`	2 concurrent workers query queued jobs via `SKIP LOCKED`	Both workers claim distinct jobs without deadlock or double claim	Pass	VERIFIED IN TEST (`test_postgres_concurrency.py`)
TC-WRK-02	`worker.fencing`	Worker attempts to commit stage after lease expiration	Raises `WorkerFencedError` ; rolls back transaction safely	Pass	VERIFIED IN TEST (`test_worker_fencing.py`)
TC-DEL-01	`retention.worker`	Execute 4-phase DPDPA deletion on existing document	Purges DB rows, deletes S3 object, writes signed HMAC certificate	Pass	VERIFIED IN TEST (`test_deletion_lifecycle.py`)
TC-REV-01	`api.analyses`	Reviewer submits sign-off (`APPROVE` / `FLAG`)	Transitions analysis status to `REVIEWED_COMPLETE`	Pass	VERIFIED IN TEST (`test_doctor_review_workflow.py`)
TC-REV-02	`api.analyses`	Patient attempts to submit review sign-off	Rejected with HTTP `403 Forbidden` (`INSUFFICIENT_ROLE`)	Pass	VERIFIED IN TEST (`test_doctor_review_workflow.py`)

4. Notable Bugs Discovered & Debugging Decisions

Bug 1: PostgreSQL Missing `patients` Table Migration (`BUG-01`)

- **Symptom:** In test suite execution on PostgreSQL, 14 integration tests failed with `relation "patients" does not exist` .
- **Root Cause:** A new `Patient` domain entity had been declared in SQLAlchemy models, but no corresponding Alembic migration script had been generated.
- **Resolution:** Authored Alembic migration 0003_add_patients_table.py safely creating the table, adding foreign key constraints, and registering the `Role.PATIENT` enum.

Bug 2: Python Dependency Conflicts in Docker Build (`BUG-02`)

- **Symptom:** Docker container build on Render failed during `pip install` with `ResolutionImpossible: google-genai depends on pydantic>=2.9.0 and httpx>=0.28.1` .
- **Root Cause:** `backend/requirements.txt` strictly pinned `httpx==0.27.0` and `pydantic==2.7.4` .
- **Resolution:** Upgraded `httpx` to `0.28.1` and loosened Pydantic constraints to `pydantic>=2.10.0,<3.0.0` in backend/requirements.txt.

Bug 3: Windows SQLite Path Invocation on Linux Container (`BUG-03`)

- **Symptom:** Render web service crashed during deployment startup with: `sqlalchemy.exc.OperationalError: (sqlite3.OperationalError) unable to open database file` .
- **Root Cause:** The developer's local `backend/.env` containing Windows path `sqlite:///c:/Users/pbhus/.../prescripto.db` had been pasted into the Render dashboard. The Linux container could not resolve drive letter `c:` .
- **Resolution:** Authored an automatic path sanitizer inside `normalize_database_url` in backend/prescripto/config/settings.py that detects Windows paths inside Linux environments and rewrites them to `sqlite:///prescripto.db` .

Bug 4: Vercel Frontend Relative API 404 Failure (`BUG-04`)

- **Symptom:** Doctor and patient login on Vercel returned error: `"PROBLEM: The page could not be found"`.
- **Root Cause:** In the frontend, `VITE_API_BASE_URL` defaulted to relative path `/api/v1`. On Vercel (static hosting), calling relative `/api/v1` hit Vercel's static router instead of the Render backend.
- **Resolution:**
- Updated web/src/api/client.ts to hardcode production fallback to `https://prescripto-2-0.onrender.com`.
- Created web/vercel.json adding rewrite rules `/* -> /index.html` for client-side routing.
- Added dynamic CORS regex `allow_origin_regex=r"https?://.*"` in FastAPI.